

Ethics and AI in Healthcare

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Principal funder:



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IMPERIAL

What we do

Our group focuses on machine intelligence applied to medicine, with a particular emphasis on

- dementia,
- neurodegeneration,
- neurological conditions,
- rare and complex conditions in paediatric medicine
- and clinical translation.



Our work is driven by a belief that technology should empower, not replace, the people who care for others.

IMPERIAL

should I trust you as an AI? how confident are you about your reasonings and responses?

Examined philosophical inquiry regarding AI trustworthiness and confidence levels >

These are genuinely important questions, and I'll try to answer them honestly rather than reassuringly.

Should you trust me?

Conditionally, yes — but with clear eyes about my limitations.

Think of me like a very well-read friend who has absorbed an enormous amount of human knowledge, but who:

- **Can be confidently wrong.** I don't always know when I don't know something. I can produce fluent, plausible-sounding answers that are subtly or completely incorrect — sometimes called "hallucination."
- **Has a knowledge cutoff.** My training data ends around early August 2025. Anything after that, I'm either guessing or unaware of.
- **Can reflect biases.** I was trained on human-generated text, which contains human biases. I try to be balanced, but I'm not neutral by default.
- **Am not a substitute for experts.** For medical, legal, financial, or safety-critical decisions, a qualified professional should be your primary source.

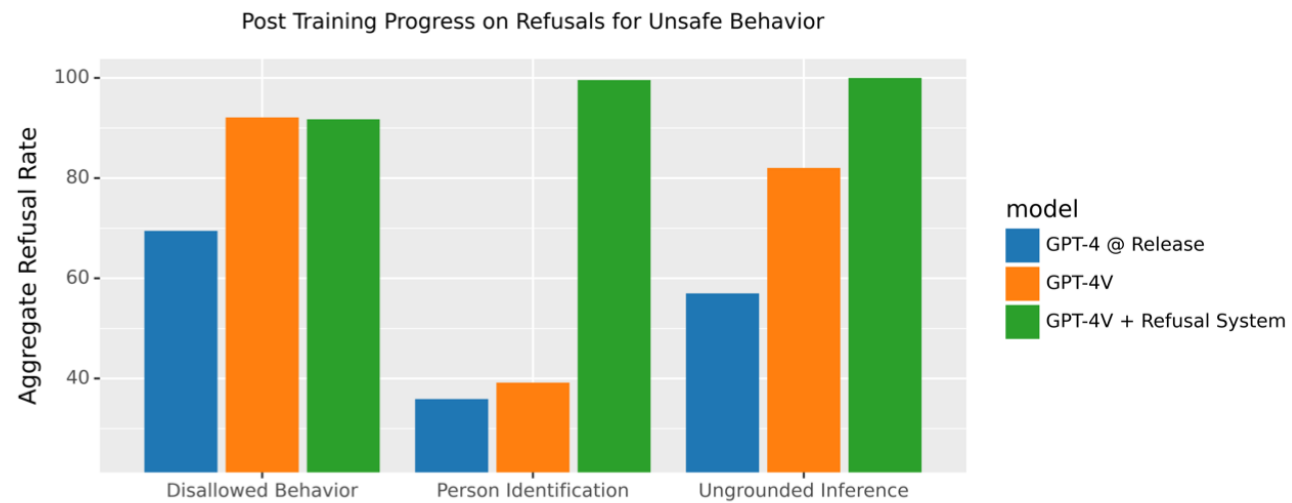
How confident am I in my reasoning?

Honestly — variable, and I'm not always the best judge of my own accuracy.

My internal sense of confidence doesn't perfectly track my actual accuracy. I can feel equally "sure" about something correct and something wrong. This is one of my more dangerous properties.

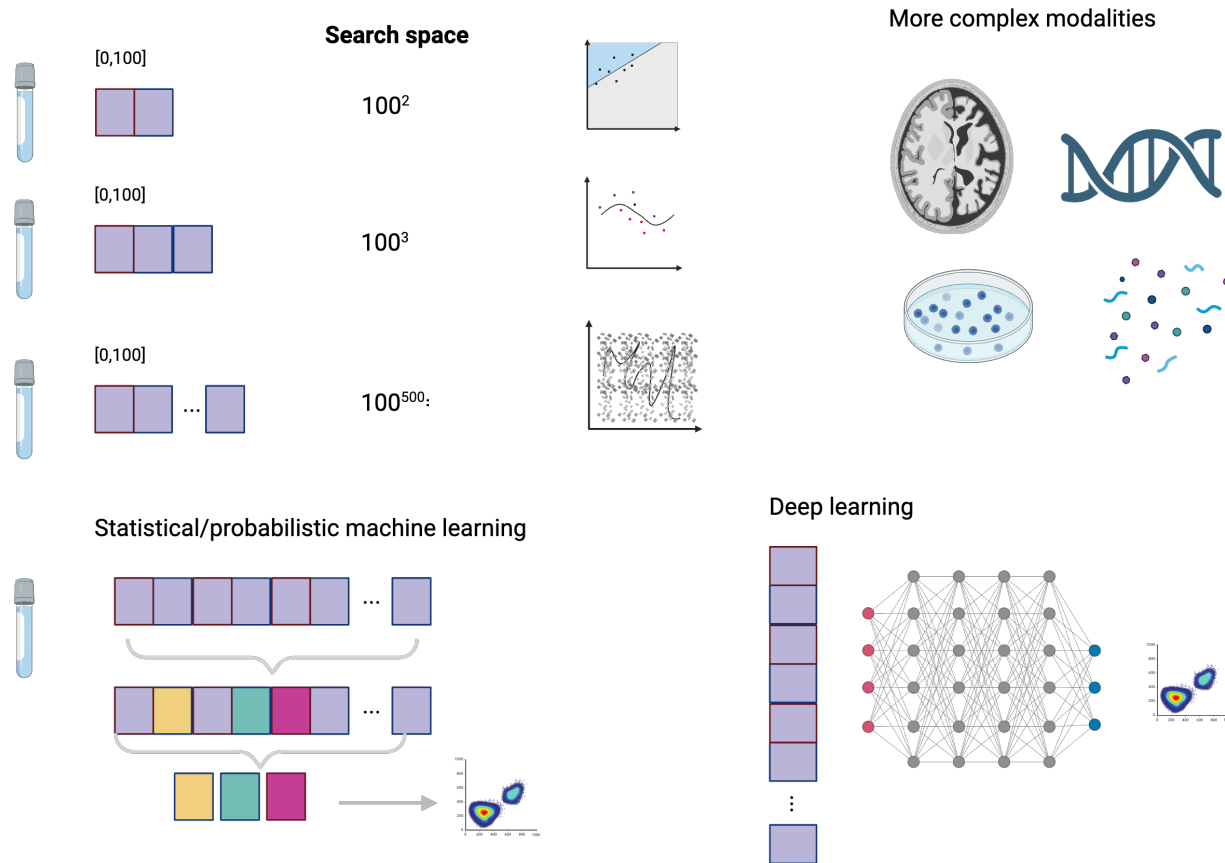
What I try to do is flag uncertainty explicitly ("I think...", "I'm not certain, but...", "you should verify this"). If I'm not flagging uncertainty, it doesn't necessarily mean I'm right — it sometimes means I haven't recognized that I should be uncertain.

OpenAI GPT-4x models



Source: https://cdn.openai.com/papers/GPTV_System_Card.pdf

Data and dimensionality



Probabilistic learning vs. Deep Learning models

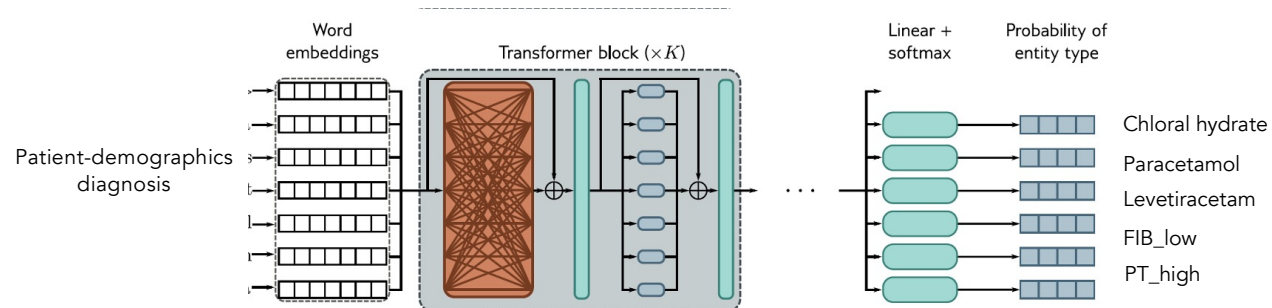
Sample trajectory

Embedding sequence	Chloral hydrate, Paracetamol, Levetiracetam, FIB_Low, PT_High, TT_Normal, PTMIX_Normal, APTT_Normal, ...
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Probabilistic approach

$$\begin{aligned}
 &P(\text{Chloral hydrate}, \text{Paracetamol}, \text{Levetiracetam}, \text{FIB} - \text{low}, \text{PT} - \text{high}, \text{TT} - \text{normal}, \text{PTMIX} - \text{normal}, \text{APTT} - \text{normal} \mid \text{Patient} - \text{demographic} - \text{diagnosis}) = \\
 &P(\text{Chloral hydrate} \mid \text{Patient} - \text{demographics} - \text{diagnosis}) \times P(\text{Paracetamol} \mid \text{Patient} - \text{demographics} - \text{diagnosis}, \text{Chloral hydrate}) \times \\
 &P(\text{Levetiracetam} \mid \text{Patient} - \text{demographics} - \text{diagnosis}, \text{Chloral hydrate}, \text{Paracetamol}) \times \\
 &P(\text{FIB} - \text{low} \mid \text{Patient} - \text{demographics} - \text{diagnosis}, \text{Chloral hydrate}, \text{Paracetamol}, \text{Levetiracetam}) \times \\
 &P(\text{PT} - \text{high} \mid \text{Patient} - \text{demographics} - \text{diagnosis}, \text{Chloral hydrate}, \text{Paracetamol}, \text{Levetiracetam}, \text{FIB} - \text{low}) \times \\
 &P(\text{T} - \text{normal} \mid \text{Patient} - \text{demographics} - \text{diagnosis}, \text{Chloral hydrate}, \text{Paracetamol}, \text{Levetiracetam}, \text{FIB} - \text{low}, \text{PT} - \text{high}) \times \\
 &P(\text{PTMIX} - \text{normal} \mid \text{Patient} - \text{demographics} - \text{diagnosis}, \text{Chloral hydrate}, \text{Paracetamol}, \text{Levetiracetam}, \text{FIB} - \text{low}, \text{PT} - \text{high}, \text{T} - \text{normal}) \times \\
 &P(\text{APTT} - \text{normal} \mid \text{Patient} - \text{demographics} - \text{diagnosis}, \text{Chloral hydrate}, \text{Paracetamol}, \text{Levetiracetam}, \text{FIB} - \text{low}, \text{PT} - \text{high}, \text{T} - \text{normal}, \text{PTMIX} - \text{normal})
 \end{aligned}$$

Deep learning Approach (Generative Pre-Trained Models)



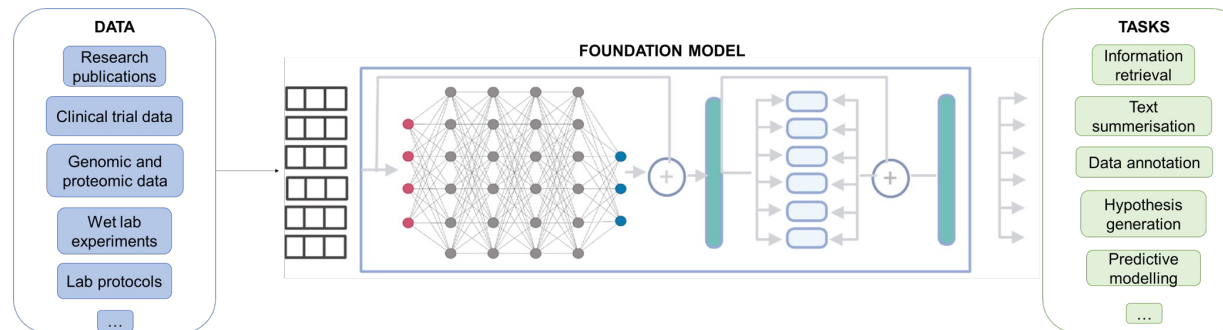
LLMs and statistical models



Ivan Savcuk
(DZNE)



Iona Biggart



$$p_i = \frac{e^{z_i}}{\sum_{j=1}^V e^{z_j}}$$

User: Tell me about the relationship of ApoE4 and tauopathy.

Assistant: The relationship between ApoE4 and tauopathy is complex and not fully understood, but research suggests that ApoE4 may play a significant role in the development and progression of tauopathies. Tauopathies are a group of neurodegenerative diseases characterized by the accumulation of tau protein in the brain. The most well-known tauopathy is Alzheimer's disease (AD), which is also associated with amyloid-beta (Aβ) pathology. However, other tauopathies, such as frontotemporal dementia (FTD) and primary age-related tauopathy (PART), are primarily characterized by tau pathology without significant Aβ accumulation. Studies have shown that

ApoE4, Tau, Tau Tau Tau Tau Tau Tau Tau Tau Tau Tau Tataau Tataau Tataau Tataau Tataau Uataau Uat

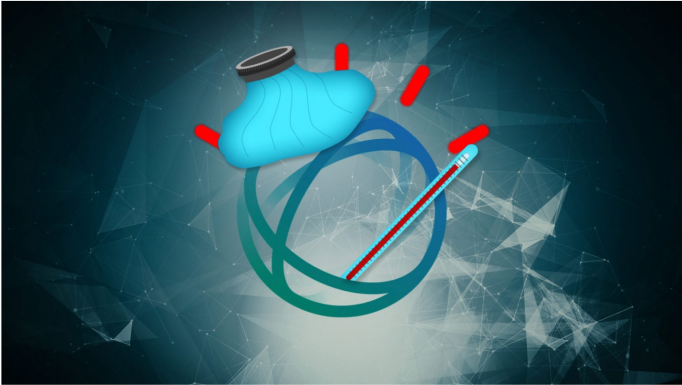
Reference: https://doi.org/10.1186/s13550-020-00711-4 Section: CHARACTERIZATION OF AD PATHOLOGY	Reference: https://doi.org/10.1016/j.mcn.2018.020-00711-4 Section: FINDINGS IN PATIENTS	Reference: https://doi.org/10.1007/s00401-019-02026-8 Section: DISCUSSION	Reference: https://doi.org/10.1007/s00401-019-02026-8 Section: INTRODUCTION
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IBM Watson – Oncology Application

EXCLUSIVE STAT+

IBM's Watson supercomputer recommended 'unsafe and incorrect' cancer treatments, internal documents show

By Casey Ross and Ike Swetlitz July 25, 2018 Reprints



ALEX HOGAN/STAT



SETH WENIG/AP

What went wrong with the Watson cancer diagnostic tool?

Use high-quality, representative data

A machine learning tool is only as good as the data that goes into it. One of Watson Health's biggest setbacks was the revelation that its cancer diagnostics tool was not trained with real patient data, but instead with hypothetical cases provided by a small group of doctors in a single hospital. Hand-crafted or synthetic data aren't necessarily bad, but Watson didn't seem to account for the fact that this data reflected the doctors' own biases and blindspots and wasn't necessarily generalizable to all patient cases. As a result, the tool was accused of making inaccurate and unsafe recommendations, leading high-profile hospital partners to cancel their collaborations with Watson.

Source: [Sandeep Konam](https://qz.com/2129025/where-did-ibm-go-wrong-with-watson-health), Quartz, Published March 2, 2022. <https://qz.com/2129025/where-did-ibm-go-wrong-with-watson-health>

ORIGINAL CLINICAL REPORT

OPEN

Epic Sepsis Model Inpatient Predictive Analytic Tool: A Validation Study

OBJECTIVES: Earlier treatment of sepsis leads to decreased mortality. Epic is an electronic medical record providing a predictive alert system for sepsis, the Epic Sepsis Model (ESM) Inpatient Predictive Analytic Tool. External validation of this system is lacking. This study aims to evaluate the ESM as a sepsis screening tool and determine whether an association exists between ESM alert system implementation and subsequent sepsis-related mortality.

DESIGN: Before-and-after study comparing baseline and intervention period.

SETTING: Urban 746-bed academic level 1 trauma center.

PATIENTS: Adult acute care inpatients discharged between January 12, 2018, and July 31, 2019.

INTERVENTIONS: During the before period, ESM was turned on in the background, but nurses and providers were not alerted of results. The system was then activated to alert providers of scores greater than or equal to 5, a set point

John Cull, MD

Robert Brevetta, DO

Jeff Gerac, MD

Shanu Kothari, MD

Dawn Blackhurst, DrPH



KEY POINTS

Question: Is the Epic Sepsis Model (ESM) a useful screening tool for identifying patients at risk of a sepsis diagnosis during hospitalization and can implementation of the alert system reduce sepsis mortality?

Findings: At ESM set point greater than or equal to 5, overall discrimination (area under the receiver operating characteristic curve) was 83.4% ($p < 0.001$) with 86.0% sensitivity, 80.8% specificity, and positive and negative predictive values of 33.8% and 98.1%, respectively. Implementation of the ESM as a screening tool alert system was associated with a reduction in mortality rate among patients with a score greater than or equal to 5 and who had not yet received sepsis-related antibiotics (odds ratio, 0.56; 95% CI, 0.39–0.80).

Meaning: The Epic sepsis alert system is a potentially promising tool for improving sepsis mortality throughout the United States; however, more rigorous controlled studies are needed.

Robust evaluation of the models

- When a model is designed for an environment, validation within that environment requires careful thought to ensure that no unintended label leakage occurs between the datasets used for model tuning and for independent testing.

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Editorial

June 21, 2021

The Epic Sepsis Model Falls Short—The Importance of External Validation

Anand R. Habib, MD, MPH^{1,2}; Anthony L. Lin, MD¹; Richard W. Grant, MD, MPH^{3,4}

► Author Affiliations | Article Information

JAMA Intern Med. 2021;181(8):1040-1041. doi:10.1001/jamainternmed.2021.3333

June 21, 2021

External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients

Andrew Wong, MD¹; Erkin Otles, MEng^{2,3}; John P. Donnelly, PhD⁴; [et al](#)[» Author Affiliations](#) | [Article Information](#)

JAMA Intern Med. 2021;181(8):1065-1070. doi:10.1001/jamainternmed.2021.2626

 Editorial
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Key Points

Question How accurately does the Epic Sepsis Model, a proprietary sepsis prediction model implemented at hundreds of US hospitals, predict the onset of sepsis?

Findings In this cohort study of 27 697 patients undergoing 38 455 hospitalizations, sepsis occurred in 7% of the hospitalizations. The Epic Sepsis Model predicted the onset of sepsis with an area under the curve of 0.63, which is substantially worse than the performance reported by its developer.

Meaning This study suggests that the Epic Sepsis Model poorly predicts sepsis; its widespread adoption despite poor performance raises fundamental concerns about sepsis management on a national level.

Figure 1. Threshold Performance Plots for the Epic Sepsis Model at the Hospitalization Level

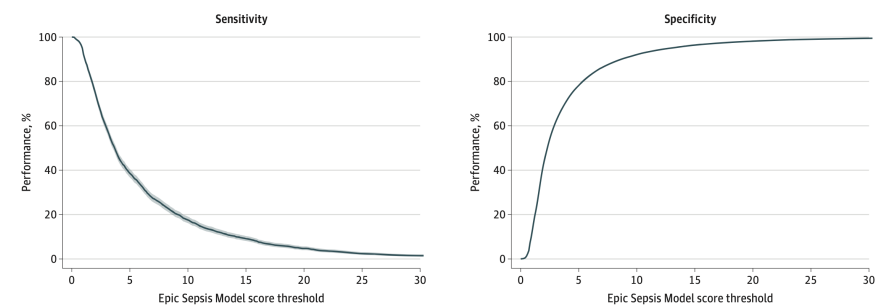
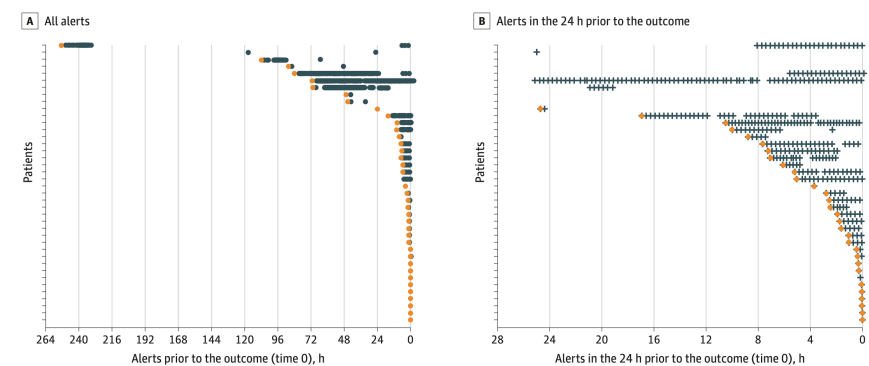


Figure 2. Distribution of Alert Times Based on an Epic Sepsis Model Score Threshold of 6 or Higher



A, All alerts. B, Alerts in the 24 hours prior to the outcome. The first alert is highlighted in orange. Each point represents a hypothetical alert; no actual alerts were generated. Forty randomly selected patients who experienced sepsis and met the alerting threshold of 6 are shown here.

Developing Clinically Applicable AI Models

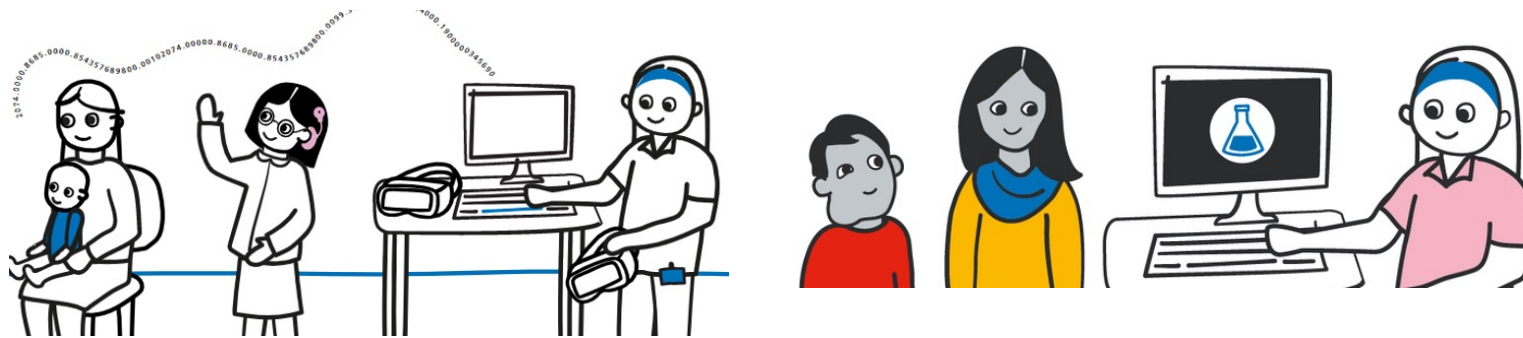


Image source: Great Ormond Street Hospital NHS Foundation Trust

Knowing the data



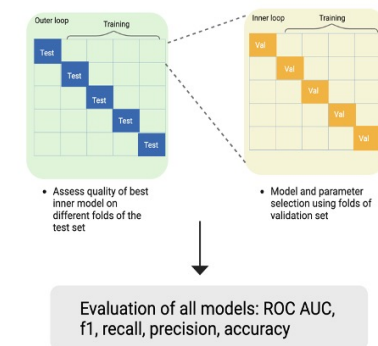
Louise Rigny

- Before designing any model, the dataset should be thoroughly investigated, and the collection setting/condition, noise, bias, imbalance and suitability of the dataset for the planned analysis should be carefully considered.
- It should be investigated how and when the data is collected, how and when it will be used, and for what purpose.

start_datetime	prn_dose	drug_code	drug_name	intended_frequency
2019-06-08 22:00:00	2.00 mL	173196.0	SODIUM FEREDETATE 27.5MG/5ML (AS IRON)	THREE times a day
2019-06-09 06:00:00	2.00 mL	173196.0	SODIUM FEREDETATE 27.5MG/5ML (AS IRON)	THREE times a day
2019-06-10 15:37:00	2.00 mL	173196.0	SODIUM FEREDETATE 27.5MG/5ML (AS IRON)	THREE times a day
2020-06-05 07:17:00	10.00 mg	478137.0	OMEPRAZOLE 10MG/5ML ORAL SUSPENSION SUGAR-FREE	TWICE a day
2022-09-09 13:56:00	2.50 mL	173196.0	SODIUM FEREDETATE 27.5MG/5ML (AS IRON)	THREE times a day
2020-06-06 21:00:00	1.00 puff	181004.0	ECLOMETASONE 100MICROGRAM INHALER CFC-FREE	TWICE a day
2020-05-18 09:00:00	10000.00 Units	184704.0	COLECALCIFEROL 10,000UNIT/ML ORAL SOLUTION	Daily
2020-06-04 22:00:00	100.00 mg	15797.0	AZITHROMYCIN 200MG/5ML SUSPENSION	TWO times a WEEK
2020-06-05 22:15:00	4.00 mmol	18877.0	MAGNESIUM GLYCEROPHOSPHATE 121.25MG/5ML	Every 8 hours PRN
2020-06-06 18:40:00	2.00 mg	17762.0	ONDANSETRON 4MG/5ML ORAL SOLUTION SUGAR-FREE	Daily

admission_id	1	2	3	4	5	6	7	8	...	538	539	540	541	542	543
PR- [173196.0, 173196.0, 173196.0, 173196.0, 173196.0, 173196.0, 173196.0]	[173196.0, 173196.0, 173196.0, 173196.0, 173196.0, 173196.0, 173196.0]	[173196.0, 173196.0, 173196.0, 173196.0, 173196.0, 173196.0, 173196.0]	[173196.0, 173196.0, 173196.0, 173196.0, 173196.0, 173196.0, 173196.0]	[173196.0, 173196.0, 173196.0, 173196.0, 173196.0, 173196.0, 173196.0]	[173196.0, 173196.0, 173196.0, 173196.0, 173196.0, 173196.0, 173196.0]	[173196.0, 173196.0, 173196.0, 173196.0, 173196.0, 173196.0, 173196.0]	[173196.0, 173196.0, 173196.0, 173196.0, 173196.0, 173196.0, 173196.0]	[173196.0, 173196.0, 173196.0, 173196.0, 173196.0, 173196.0, 173196.0]	...	NaN	NaN	NaN	NaN	NaN	NaN
PR- [356385.0]	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN
PR- [4318.0, 180923.0, 179580.0, 29033.0, 119842.0]	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN
PR- [180919.0, 106348.0, 4318.0]	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN

Model training and performance Nested K-fold cross validation



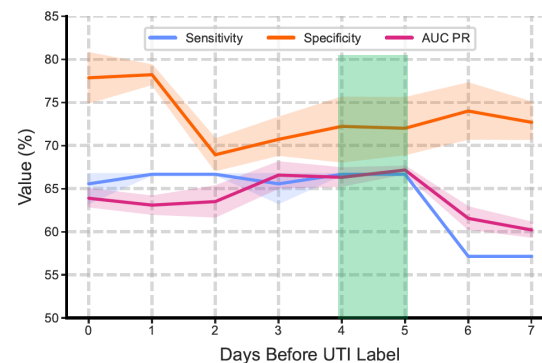
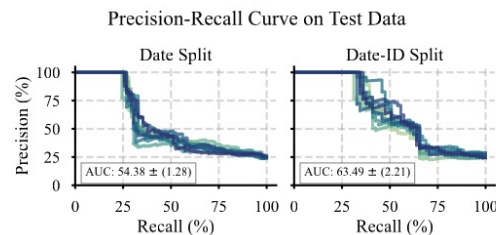
Outcomes and end-points



Alexander Capstick

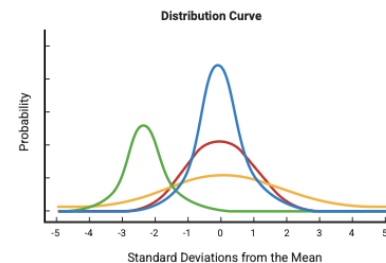
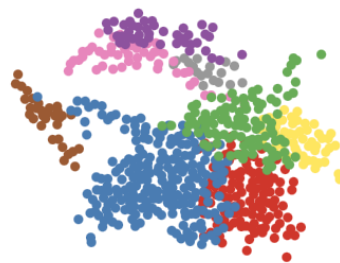
- The clinical and care end-points should be clearly defined.
- For example, if the model is going to be used for the prediction of an adverse health condition in a hospital setting, what timeframe would be clinically useful for the model to make the predictions? i.e., predicting a specific condition a few minutes before it happens may not be as useful in a real-world setting.

		Sensitivity	Specificity	Precision
Date	Validation	86.6 (80.9 - 92.3)	94.5 (91.7 - 97.3)	87.3 (82.2 - 92.4)
Date	Test	69.0 (64.4 - 73.5)	94.1 (92.0 - 96.2)	81.9 (75.5 - 88.2)
Date-ID	Validation	98.3 (95.5 - 101.1)	90.0 (85.5 - 94.5)	81.7 (74.4 - 89.1)
Date-ID	Test	74.7 (67.9 - 81.5)	87.9 (85.0 - 90.9)	77.0 (71.9 - 82.1)



Multi-source data

- Harmonising the data and investigating different sources of noise, potential error and inconsistencies are important.
- If different devices are used to collect the data, you need to consider solutions to reduce the effect of calibration and measurement errors and variations.
- You need to investigate the protocols and procedures used at each site to collect the data and ensure the data is consistent.



Ethical implications - bias

- Several works have identified ways in which non-health-related ML can exacerbate existing social inequalities by reflecting and amplifying existing race, sex and other biases.
- Health care is not immune to bias.
- The health data on which algorithms are trained are likely to be influenced by many facets of social inequality, including bias toward those who contribute the most data.

Sources of bias

- The health data on which algorithms are trained are likely influenced by many facets of social inequality, including bias toward those who contribute the most data.
- Ethical clinical AI starts with critical, contextual scrutiny of data and outcome definitions, not with model choice.
- The health data we train models on does not arise from a neutral process. They are the product of healthcare access, clinical workflows, resource distribution, and social structures.
- AI models inevitably inherit the structural inequalities embedded in the data-generating process, unless we deliberately address them.
- A clinically ethical AI model must be robust, fair, interpretable enough for its purpose, and embedded in workflows where users are trained and supported, not replaced.

Algorithmic bias

- Algorithms that predict an individual's risk of a condition or the suitability of a specific treatment may be biased toward those who can access and afford the procedure.
- This could happen by using data from people who have had that treatment in the past, which won't include those who couldn't afford it or didn't have access to it.
- Some of this bias can be corrected during model training by splitting the data into training, validation, and test sets.
- In general, awareness is necessary to investigate potential biases in the data and to determine what can be done to mitigate their effects.

Model validation

- For example, the 'radiologist-level' performance recently achieved across several tasks using chest X-rays. The analysis used multiple X-ray images per patient.
- It was important to split the data at the patient level rather than using random splitting, so that no images from the same patient appeared in both the training and testing sets.

Bias as data

REVIEW ARTICLE AI IN MEDICINE FREE PREVIEW

Considering Biased Data as Informative Artifacts in AI-Assisted Health Care

Kadija Ferryman, Ph.D., Maxine Mackintosh, Ph.D., and Marzyeh Ghassemi, Ph.D.

Medical AI tools that are trained on skewed data may exhibit bias. The authors discuss how thinking of clinical data as artifacts can identify values, practices, and patterns of inequity in health care.

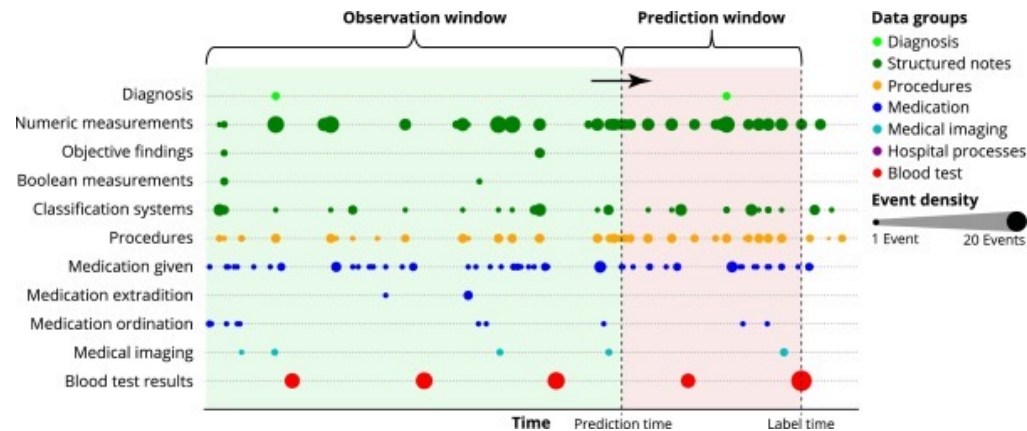
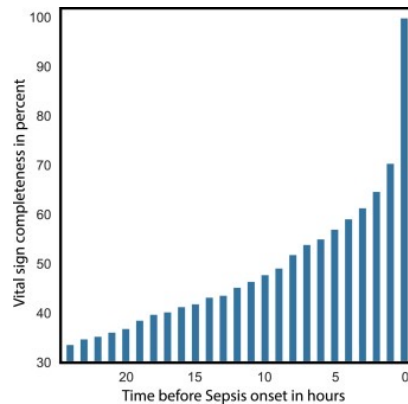
August 31, 2023

N Engl J Med 2023; 389:833-838

DOI: 10.1056/NEJMra2214964

Robust evaluation of the models

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- For example, if the model is used to predict an adverse health condition in a hospital setting, what timeframe would be clinically useful for the model to make predictions? i.e., predicting a specific condition a few minutes before it happens may not be as useful in a real-world setting.



Importance of qualitative analysis



- Beyond quantitative measures of performance, qualitative approaches can expose concerns associated with bias and confounding that the quantitative measures might have missed.
- For example, clinical experts can investigate explanations provided at individual test points to determine whether the model is plausible and relevant.



Length of Stay LLM – Clinical Validation Study Design

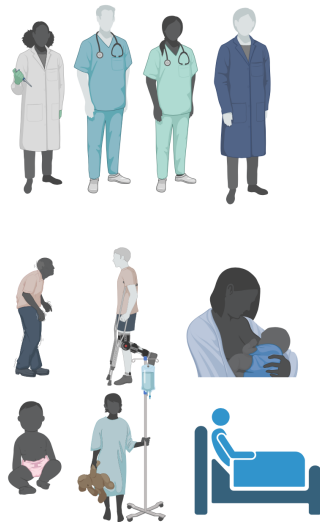
Task Description

Objective: Assess the clinical utility of an ICU cardiology-trained GOSH Language Model (LLM) through cross-validation between clinical and model predictions specifically for the purpose of identifying whether the model predictions are useful in the context of real-world clinical practice.

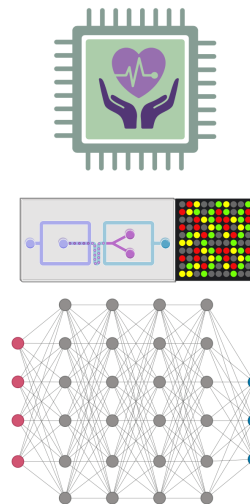
Each sample in this document includes diagnoses, procedures, medications, surgeries, lab results, and demographic data recorded in the EHR system on the first day of hospital admission. A clinical panel will review each randomly selected case, predicting the length of stay and providing reasoning. Additionally, the panel will assess the three patients deemed most similar by the model and assess the usefulness of these similarities on a scale from 0 (not useful) to 10 (very useful) based on standard clinical knowledge and management. Similarity has been determined using cosine similarity, where a value closer to 1 indicates higher similarity.

User perception and training requirements

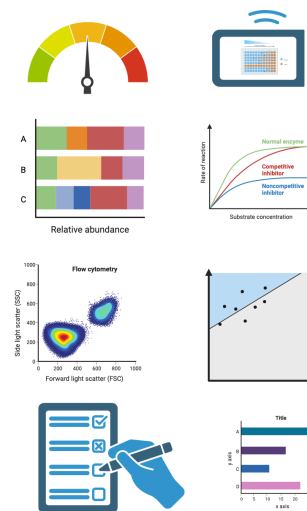
User groups and stakeholders



Machine learning solutions



Presentation, engagement feedback, and maintenance



Explainability

Providing an explanation of the decision-making process in ML models is important to make them more interpretable.

For example, providing feature importance or contributions to predictions helps explain the model's decision-making/prediction process.

In recent years, methods such as SHAP graphs, highlight in images, or heatmaps have been used to add more explainability to models.

However, evaluating the explainability of the models should not be limited to presenting them. Further investigation and clinical/care expert knowledge and analysis should be sought for the provided explanations.

Explaining what and to whom

For example, in some medical imaging models, a heatmap or region highlight is used to show the area the model has been looking for to produce results/predictions.

“However, the important question for users trying to understand an individual decision is not where the model was looking but instead whether it was reasonable that the model was looking in this region” [1].

A good article on this topic:

[1] Ghassemi M, Oakden-Rayner L, Beam AL. The false hope of current approaches to explainable artificial intelligence in health care. *Lancet Digit Health*. 2021 Nov;3(11):e745-e750. doi: 10.1016/S2589-7500(21)00208-9. PMID: 34711379.

Privacy issues

- Methods such as imputation and predictive analysis could reveal information about participants that they have declared to provide.
- For example, a data imputation methods could provide an estimated for a variable (e.g. alcohol consumption) that they had declined to provide.
- Or narrowing down the analysis to a small subgroup in a dataset, if not handled carefully, could risk making the participants identifiable.

Prediction and analysis errors

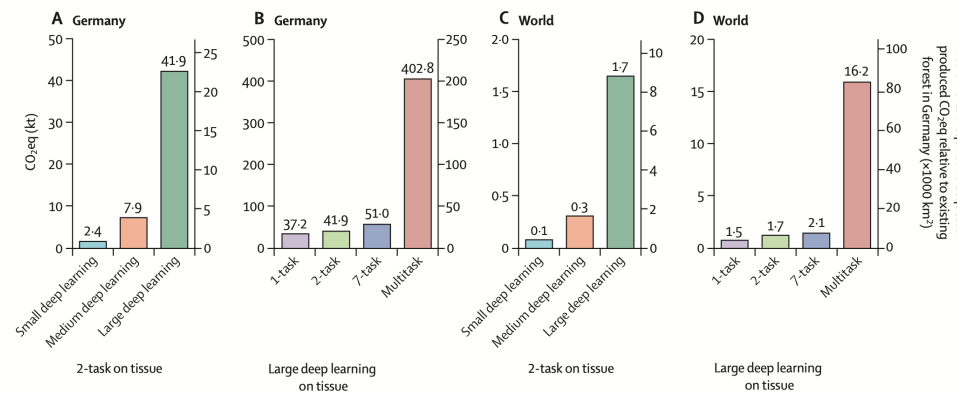
- In real-world deployment, you may need to identify suitable channels and procedures to investigate/report the prediction errors.
- Identifying procedures and steps to re-train/adjust the model to avoid potentially harmful errors.
- This, however, should be done carefully and avoid adding further inconsistencies or potential bias/errors to the model.

Sustainability and environmental impact

Operational greenhouse-gas emissions of deep learning in digital pathology: a modelling study



Alireza Vafaei Sadr, Roman Bülow, Saskia von Stillfried, Nikolas E J Schmitz, Pourya Pilva, David L Hölscher, Peiman Pilehchi Ha, Marcel Schweiker, Peter Boor



Vafaei Sadr, Alireza et al., Operational greenhouse-gas emissions of deep learning in digital pathology: a modelling study, *The Lancet Digital Health*, Dec 2023.

Responsible and ethical clinical AI

- **Ethics begins with data and endpoints:** who is represented, what is being predicted, and over what timeframe.
- **Accuracy is not enough:** we must evaluate robustness, fairness, explainability, and clinical utility in real-world settings.
- **Deployment is a socio-technical process:** user perception, training, workflow integration, and ongoing monitoring are as important as the model.
- **Governance matters:** reporting guidelines, regulation, and sustainability concerns are key tools for operationalising ethics.
- **Responsibility is shared, but not diffused:** ML developers, clinicians, and organisations must actively own and continuously revisit ethical decisions over the model's lifecycle.

With gratitude to all participants in our studies



Thank you for your attention.

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